Mathematics Bootcamp

General Equilibrium and the Social Planner

Miguel Vázquez-Carrero

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Today

- **Infinite paths:** recurrence, initial condition, and boundary at infinity
- **Household:** dated goods, shadow values, and the Euler equation
- **Competitive equilibrium:** definition, clearing, and characterization
- **Planner:** efficiency, supporting prices, and existence

Day 4 Chose a Point; Day 5 Chooses an Infinite Path

Day 4 studied a scalar choice

$$x \in D \subseteq \mathbb{R}.$$

Today the choice is an infinite sequence:

$$z = (x_0, x_1, x_2, \ldots).$$

$$\boxed{x_0} \longrightarrow \boxed{x_1} \longrightarrow \boxed{x_2} \longrightarrow \boxed{\cdots} \longrightarrow \boxed{\infty}$$

In the one-state model below, a candidate path satisfies one Euler equation at every date and one transversality condition at infinity.

Every Dated Constraint Has Its Own Multiplier

An infinite problem has one constraint at each date, so its formal Lagrangean has one multiplier at each date:

$$\max_{z=(z_t)_{t=0}^{\infty}} F(z) \quad \text{s.t.} \quad g_t(z)=0, t=0,1,\ldots, \qquad L(z;\lambda)=F(z)+\sum_{t=0}^{\infty}\lambda_t g_t(z).$$

$\boxed{g_0(z)=0}$	$\boxed{g_1(z)=0}$	$\boxed{\cdots}$	$\boxed{g_t(z)=0}$
λ_0	λ_1	$\cdots$	λ_t

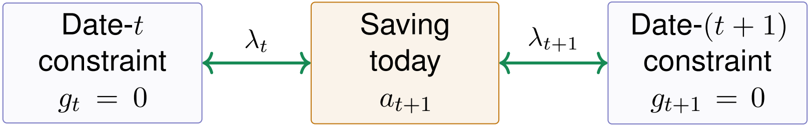
What Is Formal Here?

A general multiplier theorem for sequence spaces is beyond our scope. We use this expression to generate date-by-date candidates; feasibility, concavity, and a model-specific transversality condition do the verification.

Next Period’s State Appears in Two Neighboring Constraints

To keep the notation concrete, call the linking choice a_{t+1}: saving chosen at t and carried into $t+1$. It appears twice:

$$g_t(\dots, a_{t+1}) = 0, \qquad g_{t+1}(a_{t+1}, \dots) = 0.$$



Differentiating with respect to a_{t+1} links adjacent multipliers. Eliminating them typically produces

$$\Phi(a_t, a_{t+1}, a_{t+2}) = 0.$$

In the household problem, this recurrence will be the Euler equation.

The Euler Equation Alone Does Not Determine a Path

A recurrence alone does not determine a path:

$$\Phi(a_t, a_{t+1}, a_{t+2}) = 0, \qquad t = 0, 1, \ldots.$$

Beginning of the Path

The initial state is fixed:

a_0 is given.

End of the Path

A model-specific condition controls the tail as $t \to \infty$.

The Euler equation is local. The initial state and the tail restriction select among candidate paths; neither proves that a path exists.

In This Economy, the Recurrence Is the Euler Equation

Abstract mathematics	**Dynamic economy**
Sequence $z = (z_t)_{t=0}^{\infty}$	Paths of consumption, assets, and capital
Constraint $g_t(z) = 0$	Dated budget or resource constraint
Linking choice s_{t+1}	Saving or next-period capital
Multiplier λ_t	Shadow value of resources at date t
Recurrence	Euler equation

A Date Label Creates a Different Commodity

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label:

$$c_0 \longrightarrow c_1 \longrightarrow c_2 \longrightarrow \cdots \longrightarrow \infty$$

Different Date, Different Good

Consumption at t and $t+1$ are distinct commodities. Saving exchanges the current commodity for a future one; the interest rate will determine their relative price.

Start with the Primitives, Not the Equilibrium

Dynamic Production Economy

The environment is

$$E = \left(\mathbb{N}_0, \beta, u, f, (A_t)_{t=0}^{\infty}, k_0\right).$$

These primitives are fixed before prices and allocations are determined.

Primitives

- Dates $t = 0, 1, \ldots$
- Preferences: β and u
- Technology: f and A_t

Initial Resources

- The household owns $a_0 = k_0 > 0$.
- Labor supply is one unit each date.
- Capital does not depreciate: $\delta = 0$.

The Household Ranks Entire Consumption Paths

Lifetime Utility

For $c = (c_0, c_1, c_2, \ldots)$,

$$U(c) = \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

The discount factor β gives the relative weight on later utility. We assume that this sum is well-defined and finite on admissible paths.

Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

Strict concavity

$$u''(c_t) < 0$$

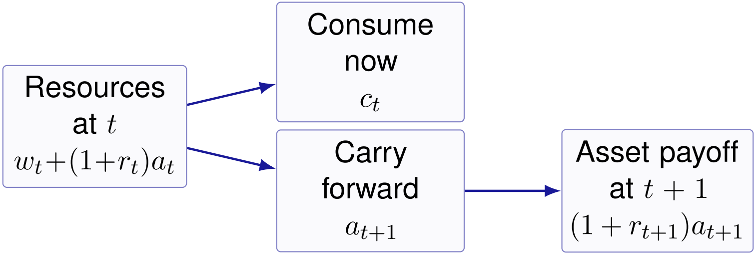
Marginal utility falls with consumption.

Saving Transfers Resources Across Dates

At date t, the household owns assets a_t, supplies one unit of labor, and receives wage and capital income.

Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$



The Household Chooses a Consumption–Saving Path

Given the price paths $(w_t, r_t)_{t=0}^{\infty}$ and initial assets a_0, the household solves

$$\max_{(c_t, a_{t+1})_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \qquad c_t \geq 0, \qquad a_{t+1} \geq 0, \qquad t = 0, 1, \ldots,$$

a_0 given.

Admissibility and Price Taking

Admissibility: $a_{t+1} \geq 0$ forbids borrowing; it is stronger than a no-Ponzi condition.

Price taking: the household treats (w_t, r_t) as given.

Each Dated Budget Constraint Has Its Own Shadow Value

At an interior path, write the formal Lagrangean

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t \left[w_t + (1+r_t)a_t - c_t - a_{t+1} \right].$$

The first-order conditions imply

$$\beta^t u'(c_t) = \lambda_t, \qquad t = 0, 1, \ldots,$$

$$\lambda_t = (1 + r_{t+1})\lambda_{t+1}, \qquad t = 0, 1, \ldots.$$

Interpretation

λ_t is the increase in lifetime utility from one additional unit of the consumption good available at date t.

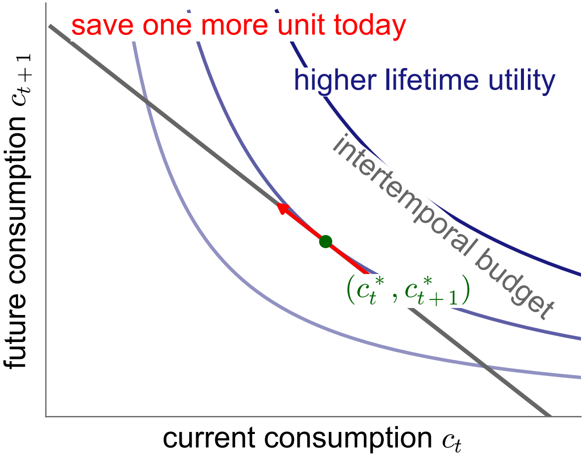
Tangency Across Dates Gives the Euler Equation

The household treats consumption at t and $t+1$ as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1+r_{t+1}}.$$

Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



Saving One More Unit: Marginal Cost Equals Marginal Benefit

Combine the two first-order conditions:

$$\boxed{u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).}$$

Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

Benefit One Date Later

Tomorrow it yields $1 + r_{t+1}$ units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

The Euler Equation Is Local; the TVC Disciplines the Tail

For the interior candidate path in this model, impose the model-specific transversality condition

$$\boxed{\lim_{t\to\infty} \beta^t u'(c_t) a_{t+1} = 0.}$$

Euler Equation

Compares the marginal cost and benefit of moving one unit between two adjacent dates.

Transversality Condition

Rules out tails that retain resources with positive shadow value.

Neither condition proves existence. Feasibility, concavity, and a separate existence argument remain necessary.

The Firm Transforms Capital and Labor into Output

In every date t, the firm rents capital k_t, hires labor ℓ_t, and produces the single final good:

$$y_t = A_t f(k_t, \ell_t) = A_t k_t^{\alpha} \ell_t^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Technology

A_t is total factor productivity. It scales output at every input pair; f remains the standard Cobb–Douglas function.

Constant Returns

Under constant returns, any finite profit-maximizing plan earns zero profit. Positive profit would scale without bound.

Throughout, $\delta = 0$: capital is fully preserved between dates.

The Firm's Problem Is Static Each Date

Normalize the final-good price to one. Given w_t and r_t, the firm solves

$$\max_{k_t,\ell_t\geq 0} \left\{ A_t f(k_t,\ell_t) - w_t\ell_t - r_t k_t \right\}.$$

At a positive-input optimum,

$$w_t = A_t \frac{\partial f(k_t,\ell_t)}{\partial \ell_t} = A_t f_2(k_t,\ell_t), \qquad r_t = A_t \frac{\partial f(k_t,\ell_t)}{\partial k_t} = A_t f_1(k_t,\ell_t).$$

Why the Firm Problem Is Static

This is the one-period problem from Day 4, repeated at every date. The firm rents inputs, produces, and sells output within date t; the household owns the capital and chooses how much to carry into the next date.

Capital Earns r_t and Returns Intact

For Cobb–Douglas production,

$$w_t = (1-\alpha)A_t k_t^{\alpha}\ell_t^{-\alpha}, \qquad r_t = \alpha A_t k_t^{\alpha-1}\ell_t^{1-\alpha}.$$

The firm pays r_t for one unit of capital. Since capital is returned intact, the household receives the gross payoff $1 + r_t$.
Constant returns and the first-order conditions imply

$$A_t f(k_t, \ell_t) = w_t \ell_t + r_t k_t,$$

so competitive profits are zero.

Physical Feasibility Contains No Prices

Allocation and Feasibility

A physical allocation is a sequence

$$z = (c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}.$$

It is **feasible** if, for every $t = 0, 1, \ldots$,

$$c_t \geq 0, \qquad k_{t+1} \geq 0, \qquad 0 \leq \ell_t \leq 1,$$

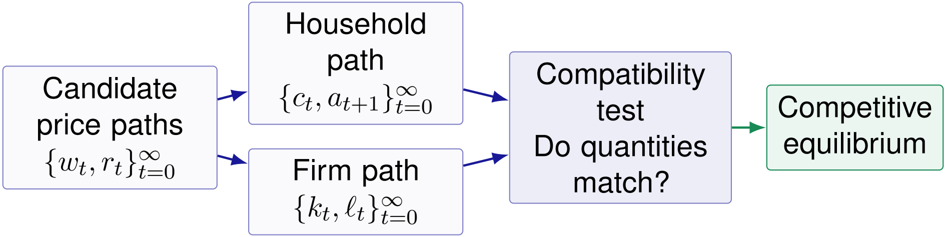
and

$$\boxed{c_t + k_{t+1} \leq A_t k_t^{\alpha} \ell_t^{1-\alpha} + k_t.}$$

Feasibility is physical and contains no prices. Monotonicity makes the resource inequality bind at an optimum.

Can Household and Firm Choices Coexist?

The household and firm have now solved separate optimization problems. General equilibrium asks whether their choices can hold at the same prices.



The Definition Is a Test, Not an Adjustment Story

It says which price-allocation paths qualify as equilibria. It does not say how an economy discovers those prices.

Optimal Choice May Be Set-Valued

Household Correspondence

At prices (w, r), define

$$H(w, r) = \{\text{household-optimal paths}\}.$$

It may contain more than one path (c, a).

Firm Correspondence

At prices (w_t, r_t), define

$$F_t(w_t, r_t) = \{\text{optimal input pairs at } t\}.$$

It may contain more than one pair (k, ℓ).

Under constant returns,

$$\text{positive profit} \implies F_t(w_t, r_t) = \varnothing, \qquad \text{zero profit} \implies F_t(w_t, r_t) \text{ may be set-valued.}$$

Equilibrium selects mutually compatible elements of H and F_t.

Define Equilibrium Before Deriving Its Equations

DEFINITION

- Names the objects that constitute an equilibrium.
- States optimization and market clearing.
- Does not require derivatives or first-order conditions.

CHARACTERIZATION

- Derives equations every equilibrium must satisfy.
- Uses first-order conditions or KKT under their hypotheses.
- Needs extra conditions for sufficiency, existence, or uniqueness.

Equilibrium Requires Optimality and Market Clearing

Competitive Equilibrium

Given a_0, a competitive equilibrium is a collection

$$((c_t, a_{t+1}), (k_t, \ell_t), (w_t, r_t))_{t=0}^{\infty}$$

with $(w_t, r_t) \in \mathbb{R}^2_{++}$ for every t, such that:

1. $(c, a) \in H(w, r)$;
2. $(k_t, \ell_t) \in F_t(w_t, r_t)$ for every t;
3. markets clear at every date:

$$a_t = k_t, \qquad \ell_t = 1, \qquad c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

Market Clearing Links Individual and Aggregate Quantities

Asset Market

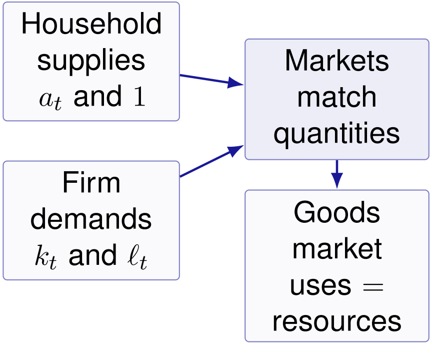
$$a_t = k_t.$$

Labor Market

$$\ell_t = 1.$$

Goods Market

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$



Why Goods-Market Clearing Follows

Suppose the household and firm optimize and the asset and labor markets clear.

(1) Substitute $a_t = k_t$ and $a_{t+1} = k_{t+1}$ into the household budget:

$$c_t + k_{t+1} = w_t + (1 + r_t)k_t.$$

(2) Subtract the capital carried into date t:

$$c_t + k_{t+1} - k_t = w_t + r_t k_t.$$

(3) Since $\ell_t = 1$ and constant returns imply zero profit,

$$w_t \ell_t + r_t k_t = A_t f(k_t, \ell_t).$$

(4) Therefore the goods market clears:

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

Firm Optimality Makes Prices Functions of Capital

To characterize equilibrium, set $A_t = 1$ and use $\ell_t = 1$.

General Technology

$$w_t = f_\ell(k_t, 1),$$
$$r_t = f_k(k_t, 1).$$

Cobb–Douglas Technology

$$w_t = (1-\alpha)k_t^{\alpha},$$
$$r_t = \alpha k_t^{\alpha-1}.$$

Firm optimality and market clearing remove prices as independent unknowns.

Feasibility Recovers Consumption from Capital

With $\ell_t = 1$ and $A_t = 1$, the resource constraint is

$$c_t + k_{t+1} = k_t^\alpha + k_t.$$

Hence,

$$\boxed{c_t = k_t^\alpha + k_t - k_{t+1}.}$$

$$\boxed{\begin{array}{c}\text{Current state}\\ k_t\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{Next capital}\\ k_{t+1}\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{Consumption } c_t\\ k_t^\alpha + k_t - k_{t+1}\end{array}}$$

Once the capital path is known, the allocation and prices can be recovered.

Three Substitutions Produce the Capital Recurrence

(1) Firm optimality replaces the return:

$$r_{t+1} = \alpha k_{t+1}^{\alpha-1}.$$

(2) Feasibility replaces consumption at both dates:

$$c_j = k_j^{\alpha} + k_j - k_{j+1}, \qquad j = t, t+1.$$

(3) Insert both expressions into the household Euler equation:

$$\boxed{\begin{aligned} u'(k_t^{\alpha} + k_t - k_{t+1}) = \beta \left[1 + \alpha k_{t+1}^{\alpha-1}\right] \\ \times u'\left(k_{t+1}^{\alpha} + k_{t+1} - k_{t+2}\right). \end{aligned}}$$

The result is a second-order difference equation in (k_t, k_{t+1}, k_{t+2}).

Solving the Euler System Does Not Prove Existence

Necessary Direction

$$\text{interior equilibrium} \quad \Longrightarrow \quad \text{capital recurrence.}$$

Converse Direction

$$\begin{array}{l} \text{capital recurrence} + \text{feasibility} + \text{initial condition} \\ + \text{transversality} + \text{sufficient optimality conditions} \end{array} \quad \Longrightarrow \quad \text{equilibrium.}$$

Neither implication proves that a solution exists or that it is unique.

Efficiency Asks Whether Any Feasible Path Is Better

Efficiency

A feasible allocation $z^* = (c_t^*, k_{t+1}^*, \ell_t^*)_{t=0}^{\infty}$ is efficient if there is no feasible $\widetilde{z}$ such that $U(\widetilde{c}) > U(c^*)$.

Efficiency uses only preferences and physical feasibility: no prices and no ownership rule. With one household, the planner selects the efficient path by maximizing the household's lifetime utility over feasible allocations.

The Planner Chooses the Best Feasible Path

With $A_t = 1$, the planner solves

$$\max_{(c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = k_t^{\alpha} \ell_t^{1-\alpha} + k_t, \qquad t = 0, 1, \dots,$$

$$c_t \geq 0, \qquad k_{t+1} \geq 0, \qquad 0 \leq \ell_t \leq 1, \qquad k_0 > 0 \text{ given.}$$

Since leisure does not enter utility and production increases in labor, $\ell_t = 1$. At an interior optimum,

$$\lim_{t \to \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

The Planner Has the Same Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t \left[k_t^{\alpha} + k_t - c_t - k_{t+1}\right].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \qquad t = 0, 1, \ldots,$$

$$\lambda_t = \lambda_{t+1} \left[\alpha k_{t+1}^{\alpha-1} + 1\right], \qquad t = 0, 1, \ldots.$$

Therefore,

$$\boxed{u'(c_t) = \beta \left[\alpha k_{t+1}^{\alpha-1} + 1\right] u'(c_{t+1}).}$$

The Market and Planner Euler Equations Coincide

Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1}.$$

$=$

Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1}.$$

Planner Euler equation:

$$u'(c_t) = \beta\left[1 + \alpha k_{t+1}^{\alpha-1}\right]u'(c_{t+1}).$$

The equality is economically powerful only because concavity and transversality make these equations sufficient.

The Market and Planner Choose the Same Allocation

Assume lifetime utility is well-defined, the relevant optima are interior, the problems are concave, and both transversality conditions hold.

Equilibrium $\Longrightarrow$ Planner

Every equilibrium allocation is a planner optimum:

$$\text{equilibrium} \Longrightarrow \text{planner optimum.}$$

Markets attain an efficient allocation.

Planner $\Longrightarrow$ Equilibrium

Every interior planner optimum can be supported by prices:

$$\text{planner optimum} \Longrightarrow \text{equilibrium.}$$

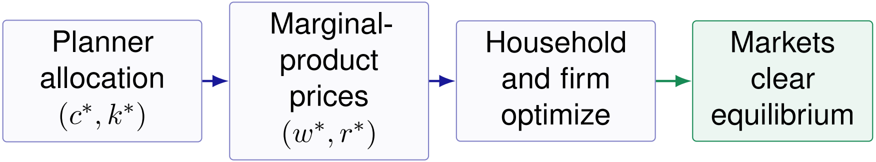
Marginal products construct the prices.

For an infinite horizon, these hypotheses are part of the theorem; the Euler equations alone are not sufficient.

Supporting Prices Decentralize the Planner Allocation

Start from an interior planner solution $(c_t^*, k_{t+1}^*)_{t=0}^{\infty}$ and define

$$w_t^* = (1-\alpha)(k_t^*)^{\alpha}, \qquad r_t^* = \alpha(k_t^*)^{\alpha-1}.$$



(1) Concavity makes the firm's marginal-product conditions sufficient.
(2) The planner Euler equation and transversality verify household optimality.
(3) Set $a_t^* = k_t^*$ and $\ell_t^* = 1$; planner feasibility is goods clearing.

Infinite-Horizon Existence Requires a Separate Theorem

This is a problem over sequences, so the finite-dimensional Extreme Value Theorem does not apply directly. A rigorous existence proof must specify:

(1) a topology on the path space;
(2) compactness or sequential compactness of the feasible set;
(3) continuity and uniform control of the discounted tail.

Two Separate Conclusions

$$\text{planner solution exists} + \text{supporting-price theorem} \Longrightarrow \text{equilibrium exists},$$

$$\begin{array}{c} \text{planner solution is unique,} \\ \text{equilibrium} \Longrightarrow \text{planner solution} \end{array} \Longrightarrow \begin{array}{c} \text{at most one} \\ \text{equilibrium allocation.} \end{array}$$

Definition Comes First; Characterization Makes It Usable

General equilibrium is optimization made mutually consistent.